

JEE MAIN REVISION MASTER: COORDINATION COMPOUNDS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 319 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Coordination Compounds - Core Application	319 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Coordination Compounds - Core Application (Very High Yield — 319 PYQs)

Core Principle: Key problem pattern in Coordination Compounds.

■ ■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Which of the following sequences of hybridisation, geometry and magnetic nature are correct for the given coordination compounds? A. $[\text{NiCl}_4]^{2-}$..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "Identify the correct statements from the following A. $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$ is the most stable complex among $[\text{Fe}(\text{OH})_6]^{3+}$, $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Match the LIST-I with LIST-II List-I Complex ion List-II Calculated spin only magnetic moment (BM) A. $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$ I. 3..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 21st January Evening Shift (Q#1)

Given below are two statements : Statement I : Crystal Field Stabilization Energy (CFSE) of $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$ is greater than that of $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$. Statement II : Potassium ferricyanide has a greater spin-only magnetic moment than sodium ferrocyanide. In the light of the above statements, choose the correct answer from the options given below :

- (A) Statement I is true but Statement II is false
- (B) Both Statement I and Statement II are true **[Correct]**
- (C) Both Statement I and Statement II are false
- (D) Statement I is false but Statement II is true

Correct Answer: B

Solution Analysis: Statement I: In $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$, Cr^{2+} is d^4 and with H_2O (weak field) it is high spin : $t_{2g}^3 e_g^1$. $\text{CFSE} = 3(-0.4\Delta_o) + 1(+0.6\Delta_o) = -0.6\Delta_o$ In $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$, Mn^{2+} is d^5 high spin: $t_{2g}^3 e_g^2$. $\text{CFSE} = 3(-0.4\Delta_o) + 2(+0.6\Delta_o) = 0$ So CFSE of the chromium complex is greater (more stabilizing). Statement I is true. Statement II: $\text{K}_3[\text{Fe}(\text{CN})_6]$: Fe^{3+} is d^5 ; CN^- is strong field $\rightarrow$ low spin t_{2g}^5 , 1 unpaired electron $\rightarrow \mu_{\text{so}} = \sqrt{1(1+2)} = \sqrt{3}$ BM. $\text{Na}_4[\text{Fe}(\text{CN})_6]$: Fe^{2+} is d^6 ; CN^- is strong field $\rightarrow$ low spin t_{2g}^6 , 0 unpaired electron $\rightarrow \mu_{\text{so}} = 0$ BM. Statement II is true.